



Research Paper

Numerical study of laminar flow and heat transfer in microchannel heat sink with offset ribs on sidewalls

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HIGHLIGHTS

- Three-dimensional model for laminar flow and heat transfer characteristics.
- Heat transfer performance for microchannel heat sink with offset ribs on sidewalls.
- Nusselt number, friction factor and performance evaluation criteria have been shown.

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ABSTRACT

A numerical investigation has been carried out to examine the characteristics of laminar flow and heat transfer in microchannel heat sink with offset ribs on sidewalls. The three-dimensional equations considering entrance effect, conjugate heat transfer, viscous heating and temperature-dependent properties are solved for the fluid flow and heat transfer in the microchannel heat sink. Five different shapes of offset ribs are designed, including rectangular, backward triangular, isosceles triangular, forward triangular and semicircular. Results show that the offset ribs result in significant heat transfer enhancement and higher pressure drop. Depending on the different offset ribs and Reynolds number ($190 \leq Re \leq 838$) studied in the present work, Nusselt number and friction factor for the microchannel heat sink with offset ribs are 1.42–1.95 and 1.93–4.57 times higher than those for the smooth one, leading to performance evaluation criteria of 1.02–1.48. Further, as a consequence of significant pressure drop, the microchannel heat sink with offset ribs gradually loses its advantage as an effective heat transfer enhancement method at higher Reynolds number.

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1. Introduction

The pioneering work by Tuckerman and Pease [1] opened the door for further investigations in the use of microchannels for high heat flux dissipation devices. Since then, much effort has been dedicated to the capabilities of microchannel heat sink to remove heat generated by electronic chips. Steinke and Kandlikar [2] studied the friction factor for single-phase liquids in microchannels and performed an in-depth comparison of experimental data to identify the discrepancies. Rosa and Karayiannis [3] reviewed the experimental and numerical results for microscale single-phase heat transfer and concluded that heat transfer in microchannels could be satisfactorily described by theory and correlations for macroscale, but indicated that the scaling effects should be accounted for, including entrance effect, conjugate heat transfer, viscous heating,

temperature-dependent properties, etc. Adham et al. [4] investigated the heat transfer and hydrodynamic performance of microchannel heat sinks and reviewed the methodologies used to analyze and optimize the overall performance of microchannel systems with regard to channel geometry, flow conditions and coolant used.

More recently, with increasing and ongoing emphasis on size reduction and strict temperature limitation in microscale thermal systems, microchannel heat sink with passive microstructures is considered to be efficient means to meet the demand. Xu et al. [5,6] experimentally and numerically studied heat transfer with laminar flow in a microchannel heat sink comprising parallel longitudinal microchannels with several transverse microchambers. They found that the heat sink could significantly reduce the pressure drop and enhance heat transfer, due to the shortened effective flow length. Chai et al. [7] studied the pressure drop and heat transfer characteristics of interrupted microchannel heat sinks with rectangular ribs in transverse chambers. Based on performance evaluation criteria, they obtained the optimal dimensions and locations of the

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rectangular ribs. Cheng [8] numerically simulated heat transfer in a stacked two-layer microchannel heat sink with passive microstructures. They found that the stacked microchannel heat sink had better performance than the smooth one. Chen et al. [9] numerically analyzed laminar flow and heat transfer in microchannel heat sink with rough surfaces and found that the heat transfer coefficient increased almost linearly with Reynolds number and was larger than the smooth one. Sui et al. [10,11] and Mohammed et al. [12] experimentally and numerically studied laminar flow and heat transfer in wavy microchannel heat sink and found that it had better heat transfer performance than the smooth one. Ghaedamini et al. [13] numerically studied the developing forced convection in converging–diverging microchannel heat sink. Based on the performance factor introduced, the superiority of converging–diverging design showed itself at higher Reynolds number for which higher performance of up to 20% was observed. Chai et al. [14] experimentally and numerically studied the heat transfer performance of microchannel heat sink with periodic expansion–contraction cross-sections. They found that the maximum of performance evaluation criteria was about 1.8 at Reynolds number ranged from 147 to 752. Danish et al. [15] and Xia et al. [16,17] analyzed the effect of geometric parameters on pressure drop and heat transfer in the microchannel heat sink with grooved structures or reentrant cavities, and obtained the optimal geometric parameters based upon performance evaluation criteria. Foong et al. [18] and Liu et al. [19] experimentally and numerically investigated the fluid flow and heat transfer characteristics in microchannel heat sink with longitudinal vortex generators (LVGs) and found that the microchannel heat sink with LVGs could enhance heat transfer but have larger pressure drop, compared with the smooth one. Promvonge et al. [20] carried out a numerical investigation to examine laminar flow and heat transfer characteristics in a three-dimensional isothermal wall square channel with 45°-angled baffles. They found that the channel with 45°-angled baffles showed the best performance evaluation criteria of 2.6 at Reynolds number ranged from 100 to 1000.

Based on the passive heat transfer enhancement methods mentioned above, the offset ribs are mounted on the two opposite channel sidewalls for microchannel heat sink with the purposes as follows. One is to develop the converging–diverging or wavy microchannels, which have played an important role in fluid mixing and heat transfer as indicated by Ghaedamini et al. [13] and Sui et al. [10,11]. Two is to work as micromixers or baffles, which can enhance the mixing of cold and hot fluids and the heat transfer as indicated by Cheng [8] and Promvonge et al. [20]. The last is to have the advantage of simple structure, easy manufacturing by micro machining method, low cost, and higher reliability than the active measures. In the present investigation, numerical simulation is conducted for laminar flow and heat transfer in the microchannel heat sink with offset ribs on sidewalls. Five different shapes of offset ribs are designed, including rectangular, backward triangular, isosceles triangular, forward triangular and semicircular. Since the offset ribs may play a fundamental role in establishing flow structure, and therefore heat transfer and pressure drop performances, the present work focuses on these characteristics and the effects of the different offset ribs.

2. Computational method

2.1. Conservation equations

A three-dimensional, incompressible, steady state laminar flow model is designed. As indicated by Zhang et al. [21], the conjugate effect of wall conduction and fluid axial conduction can be important for simultaneously developing laminar flow and heat transfer in microchannels, thus the thermal conduction along flow direc-

tion and viscous dissipation are considered in this model. The governing equations in the Cartesian tensor form are:

Continuity equation

$$\frac{\partial}{\partial x_i}(\rho_f u_i) = 0 \quad (1)$$

Momentum equation

$$\frac{\partial}{\partial x_i}(\rho_f u_i u_j) = -\frac{\partial p}{\partial x_j} + \frac{\partial}{\partial x_i} \left[\mu_f \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right) \right] \quad (2)$$

Energy equation

$$\frac{\partial}{\partial x_i}(\rho_f u_i c_{pf} T) = \frac{\partial}{\partial x_i} \left(k_f \frac{\partial T}{\partial x_i} \right) + \mu_f \left[2 \left(\frac{\partial u_i}{\partial x_i} \right)^2 + \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right)^2 \right] \quad (3)$$

$$\text{For the solid } \frac{\partial}{\partial x_i} \left(k_s \frac{\partial T}{\partial x_i} \right) = 0 \quad (4)$$

where ρ is density, μ is dynamic viscosity, c_p is specific heat capacity, k is thermal conductivity, x_1 , x_2 and x_3 are x , y and z coordinates, respectively, as shown in Fig. 1. Subscripts f and s refer to fluid and solid, respectively.

2.2. Computational domain and boundary conditions

Figure 1 illustrates the computational domain, corresponding coordinate system and key notations used. The fluid velocity at the microchannel inlet is assumed uniform

$$x=0: u_f = u_{in} \quad \text{and} \quad T_f = T_{in} \quad (5)$$

where u_{in} and T_{in} are the given fluid velocity and temperature at the channel inlet. At the channel outlet, a pressure-outlet boundary condition is applied

$$x=L: p_f = p_{out} \quad (6)$$

where p_{out} is the given pressure. For the inner wall/ fluid contact surface,

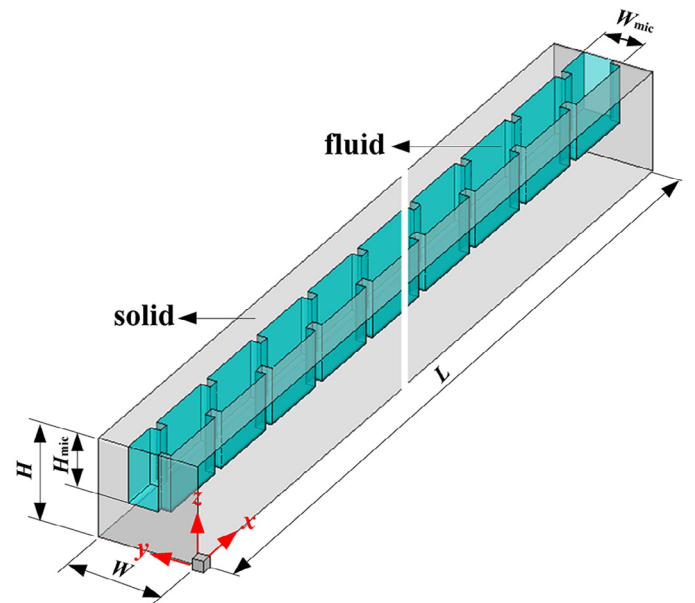


Fig. 1. Structure of microchannel heat sink with offset ribs on sidewalls.

$$u = v = w = 0, \quad -k_s \frac{\partial T_s}{\partial n} = -k_f \frac{\partial T_f}{\partial n} \quad (7)$$

where n is the local coordinate normal to the wall. At the base of computational domain,

$$z = 0: -k_s \frac{\partial T_s}{\partial z} = q_w \quad (8)$$

At the two sides of the computational domain,

$$y = 0, \quad \frac{\partial T_s}{\partial y} = 0 \quad (9a)$$

$$y = W, \quad \frac{\partial T_s}{\partial y} = 0 \quad (9b)$$

For other surfaces,

$$\frac{\partial T_f}{\partial x} = 0 \quad \text{or} \quad \frac{\partial T_s}{\partial x} = 0 \quad (10)$$

The parameters considered are in the following ranges: $u_{in} = 1 - 8 \text{ m}\cdot\text{s}^{-1}$, $T_{in} = 293 \text{ K}$, $p_{out} = 0$ (gauge pressure) and $q_w = 10^6 \text{ W}\cdot\text{m}^{-2}$. The fluid and solid are water and silicon, respectively. The thermo-physical properties of water including ρ_f , μ_f , c_{pf} and k_f depend on temperatures in accordance to Incropera [22] as shown in Eqs. (11)–(14).

$$\rho_f = \frac{999.84 + 18.225T - 7.92 \times 10^{-3}T^2 - 5.545 \times 10^{-5}T^3 + 1.498 \times 10^{-7}T^4 - 3.933 \times 10^{-10}T^5}{1 + 1.816 \times 10^{-2}T} \quad (11)$$

$$\mu_f = 2.414 \times 10^{-5} \times 10^{\frac{247.8}{T-140}} \quad (12)$$

$$c_{pf} = 8958.9 - 40.535T + 0.11243T^2 - 1.014 \times 10^{-4}T^3 \quad (13)$$

$$k_f = -0.58166 + 6.3556 \times 10^{-3}T - 7.964 \times 10^{-6}T^2 \quad (14)$$

where the unit of T is °C in Eq. (11) and K in Eqs. (12), (13) and (14). The thermal conductivity of silicon k_s is set as a constant of $148 \text{ W}\cdot\text{m}^{-1} \text{ K}^{-1}$ in the computations. The length (L), width (W) and height (H) of the computational domain are 10 mm, 0.25 mm and 0.35 mm, respectively. The width (W_{mic}) and height (H_{mic}) in the microchannel region without ribs are 0.1 mm and 0.2 mm, respectively. Figure 2 shows the detailed dimensions of the offset ribs. To determine the effect of offset ribs on heat transfer and pressure drop, five different shapes are designed: rectangular, backward triangular, isosceles triangular, forward triangular and semicircular. The five microchannel heat sinks are named for short as No.1, No.2, No.3, No.4 and No.5 as shown in Fig. 2. All the ribs are 0.05 mm in x -direction, 0.025 mm in y -direction for the maximum, 0.2 mm in z -direction and 0.4 mm in the spacing. To determine the changes of flow structure, heat transfer and pressure drop performances, a smooth microchannel heat sink without offset ribs is chosen and for short named as No.6.

2.3. Solution methods and convergence criteria

The simulations are performed using the ANSYS FLUENT 12.0 software. The SIMPLEC algorithm is applied to solve the governing differential equations for the velocity, pressure and temperature fields in the microchannels. For the microchannels with offset ribs on side-walls, an unstructured mesh based on rectangular grid elements is

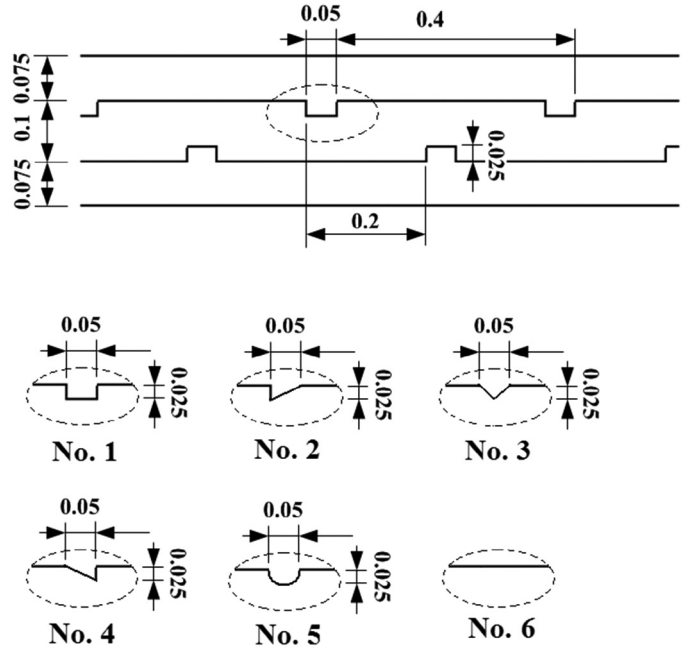


Fig. 2. Structure and dimensions of offset ribs (dimensions in mm).

utilized. A non-uniform arrangement with a large number of grids is arranged near the offset ribs. The solutions are considered to be converged when the normalized residual values are less than 10^{-5} for all the variables. The numerical code is verified in a number of ways to ensure the validity. For every microchannel heat sink, a grid independence test is conducted using several different mesh sizes. For example, as Reynolds number is 508, the deviations of average friction factor for heat sink No.1 using 0.268, 0.419 and 0.655 million grids from that of 1.023 million grids are 12.5 %, 6.7 %, 2.6 % and 0.9 % respectively, and the deviations of Nusselt number using 0.268, 0.419 and 0.655 million grids from that of 1.023 million grids are 11.6 %, 6.4 %, 2.9 % and 0.9 % respectively. Thus 0.655 million grids is used for heat sink No.1. Adopting the same test method, 0.675, 0.681, 0.675 and 0.669 million grids are used for heat sinks No.2, No.3, No.4 and No.5, respectively. Figure 3 shows the unstructured grids in the x - y plane ($z = 0.25 \text{ mm}$) of the computational domains.

2.4. Data acquisition

Parameters characterizing the fluid flow and heat transfer in the microchannels are of interest in the present work.

The Reynolds number is defined as

$$Re = \frac{\rho_f u_m D_h}{\mu_f} \quad (15)$$

where ρ_f is the volume-average fluid density, u_m is the average flow velocity in the smooth channel section, D_h is the hydraulic diameter calculated based on the channel section without offset ribs, μ_f is the mass-average fluid dynamic viscosity.

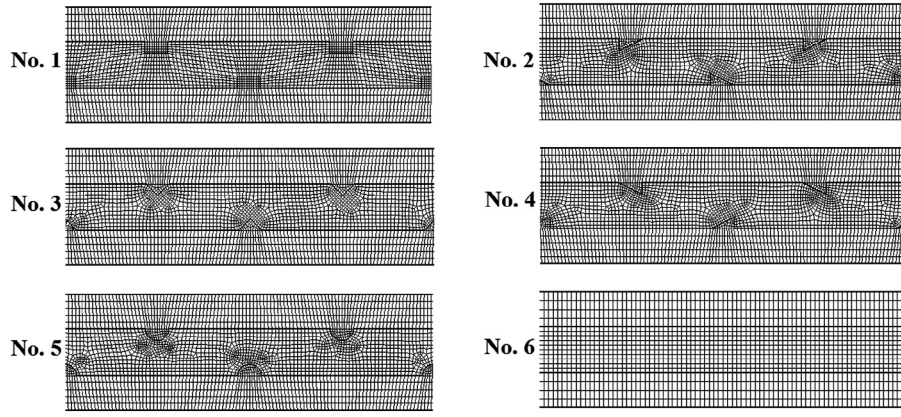


Fig. 3. Grid generations in the x-y plane ($z = 0.25$ mm).

The average Fanning friction factor is given by

$$\bar{f} = \frac{\Delta p D_h}{2 \rho_f L u_m^2} \quad (16)$$

where L is the length of the microchannel and Δp is the pressure drop across the length of the microchannel $\Delta p = \bar{p}_{in} - p_{out}$, $\bar{p}_{in}$ is the mass-weighted average inlet pressure.

The average heat transfer coefficient and Nusselt number are given by

$$\bar{h} = \frac{q_w A_w}{A_{con} (\bar{T}_w - \bar{T}_f)} \quad (17)$$

$$\overline{Nu} = \frac{\bar{h} D_h}{k_f} \quad (18)$$

where A_w is the silicon base area, A_{con} is the inner wall/ fluid contact surface area, k_f is the mass-average fluid thermal conductivity, $\bar{T}_w$ is the area-weighted temperature of the silicon base, $\bar{T}_f$ is the mass-average temperature of water in microchannels.

In order to access the enhanced heat transfer surfaces in heat exchanger design, Webb [23] outlined detailed procedures to calculate the performance improvement and to select the optimum surface geometry. For the microchannel heat sink with offset ribs on sidewalls, the performance evaluation criteria (PEC) is defined as the ratio of the heat transfer coefficient ($\bar{h}$) to the smooth microchannel heat sink ($\bar{h}_0$) at an equal pumping power. For a constant pumping power,

$$(\dot{V} \Delta p)_0 = (\dot{V} \Delta p) \quad (19)$$

The relationship between the average friction factor and Reynolds number [24] can be expressed as

$$\bar{f}_0 Re_0^3 = \bar{f} Re^3 \quad (20)$$

$$Re_0 = Re (\bar{f} / \bar{f}_0)^{1/3} \quad (21)$$

Therefore, the PEC is given by

$$PEC = \frac{\bar{h}}{\bar{h}_0} \bigg|_{pp} = \frac{\overline{Nu}}{\overline{Nu}_0} \bigg|_{pp} = \frac{\overline{Nu} / \overline{Nu}_0}{(\bar{f} / \bar{f}_0)^{1/3}} \quad (22)$$

where $\overline{Nu}$ and $\bar{f}_0$ respectively stand for Nusselt number and friction factor of the smooth microchannel heat sink.

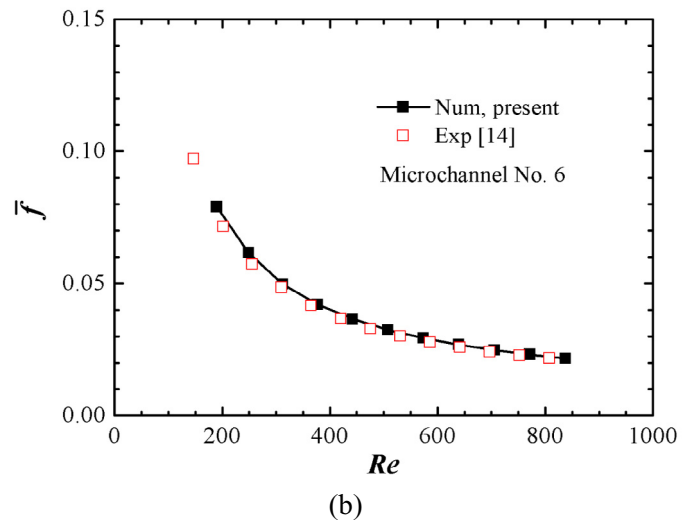
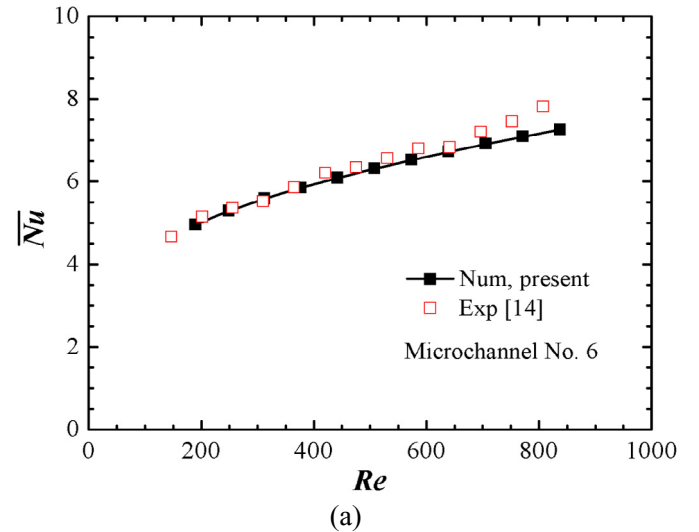


Fig. 4. Comparison of numerical results with experimental data of Chai et al. [14] for the smooth microchannel heat sink. (a) $\overline{Nu}$; (b) $\bar{f}$.

3. Results and discussion

3.1. Validation for the smooth microchannel heat sink

Validations of heat transfer and friction factor of the smooth microchannel heat sink are performed by comparing with the experimental values obtained from Chai et al. [14] as shown in Fig. 4a and b. Note that for Nusselt number and friction factor, the comparisons are found generally to be in good agreement. Having completed the grid independence study, and validation of the modeling scheme by comparison with experimental results, the next sections give the results of flow structure, heat transfer and pressure drop in the microchannel heat sink with offset ribs on sidewalls.

3.2. Flow structure

Because the flow structure plays a crucial role in heat sink performance, it is necessary to understand the flow structure in the microchannel with offset ribs. The flow structures in the microchannels with offset ribs are presented in Fig. 5a and b. Here the streamlines are respectively given in the x - y plane ($z = 0.25$ mm) and y - z plane ($x = 5.3$ mm) at $Re = 508$. It is noted that the presence of offset ribs makes the velocity distribution for laminar flow structure different from the smooth channel. The most notable characteristics include interruption of boundary layer formation, recirculation or establishment of secondary flows. As the mainstream flow passes over the offset ribs, the flow deflects in the

upstream of the rib and a recirculation is formed in the downstream of the rib, which can thin the thermal boundary layer and enhance the mixing of cold and hot fluids [20]. Comparing the microchannels No.2, No.3 and No.4 suggests that the extent of the recirculation is related to the separation point of the mainstream at the rib tip. Comparing the microchannels No.1 and No.2, No.3 and No.5, indicates that the geometry of the sidewall behind separation point has an influence on the development of recirculation.

Figure 6 shows the velocity distributions in the x - y plane ($z = 0.25$ mm). In the smooth channel, the maximum velocity always appears near the channel center. For the microchannel with offset ribs, the maximum velocity moves to the other sidewall without ribs. As shown in Fig. 6, the velocity is very small in the recirculation zone. Due to the small velocity there, a laminar stagnation zone may form which can decrease heat transfer mainly attributing to the significantly lower conduction level [25]. Further, the velocity gradient in the y direction suggests the hot and cold water mixing level. The smaller the velocity difference in the y -direction is, the better the mixing between the hot water near the wall and the cold water in the microchannel center. As shown in Fig. 6, the velocity difference in the y -direction in the microchannel No.1 is a little larger than the other microchannels with offset ribs. Further observation shows that there is a very small laminar stagnation zone in the upstream of the ribs which more easily occurs in microchannels No.1 and No.2. The velocity components in x -direction in the centerline of the microchannels ($y = 0.125$ mm, $z = 0.25$ mm) are displayed in Fig. 7. It is interesting to note that the flow direction changes periodically due to the periodical obstruction of the offset ribs, and

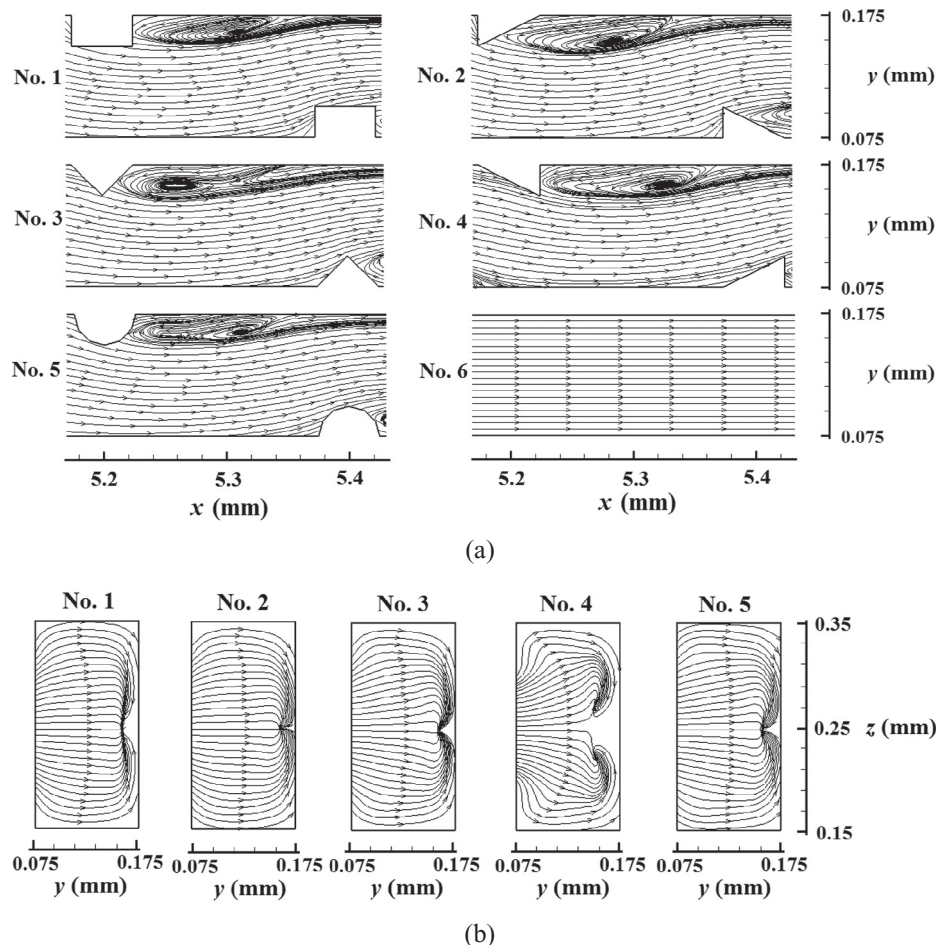


Fig. 5. Streamlines in the microchannels at $Re = 508$. (a) in the x - y plane ($z = 0.25$ mm); (b) in the y - z plane ($x = 5.3$ mm).

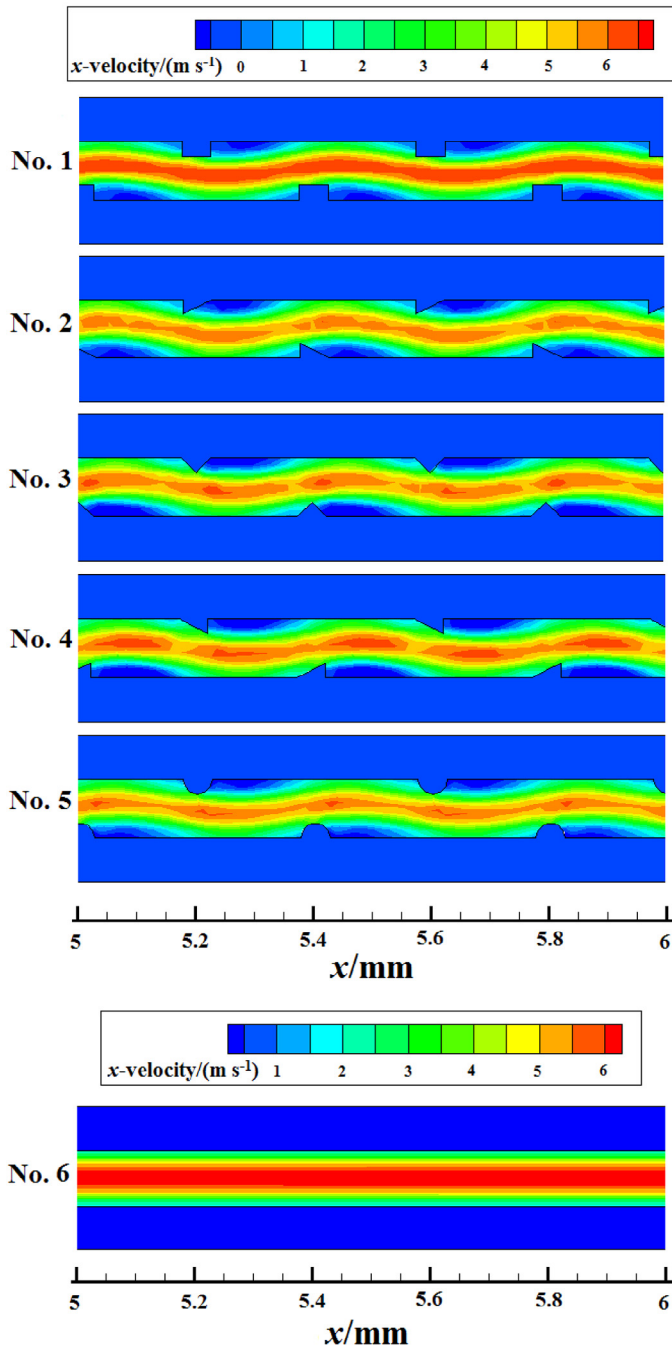


Fig. 6. Velocity distributions in the microchannels ($Re = 508$, $z = 0.25$ mm).

the velocity component in x-direction for microchannel No.1 is significantly larger but the amplitude of variation is smaller than the other microchannels with offset ribs.

3.3. Heat transfer

Figures 8 and 9 respectively show the temperature of the silicon base ($\bar{T}_w$) and average Nusselt number ($\bar{Nu}$) with Reynolds number (Re) ranging from 190 to 838. The lower $\bar{T}_w$ and higher $\bar{Nu}$ for the newly proposed heat sink show that the offset ribs can give rise to significant heat transfer enhancement. It can be seen that for the microchannel heat sinks with offset ribs, $\bar{Nu}$ rise rapidly at a smaller Re and the increase rate slows down with increase of Re . Furthermore, different offset ribs lead to different heat transfer performance,

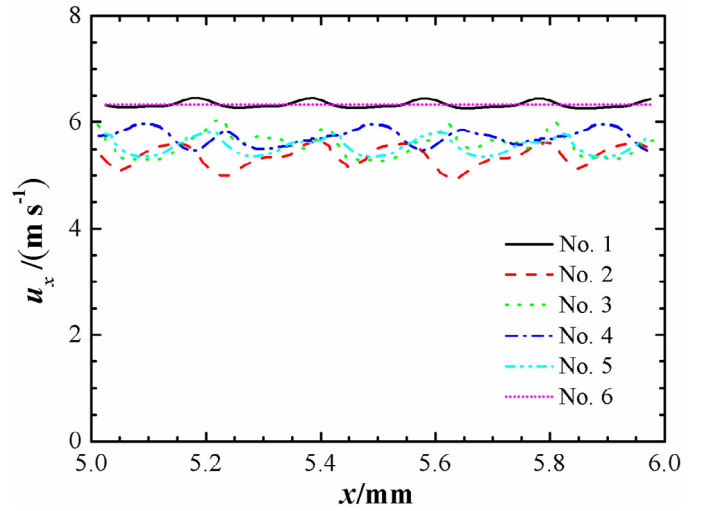


Fig. 7. Velocity component in x-direction at the centerline of microchannels ($Re = 508$, $y = 0.125$ mm, $z = 0.25$ mm).

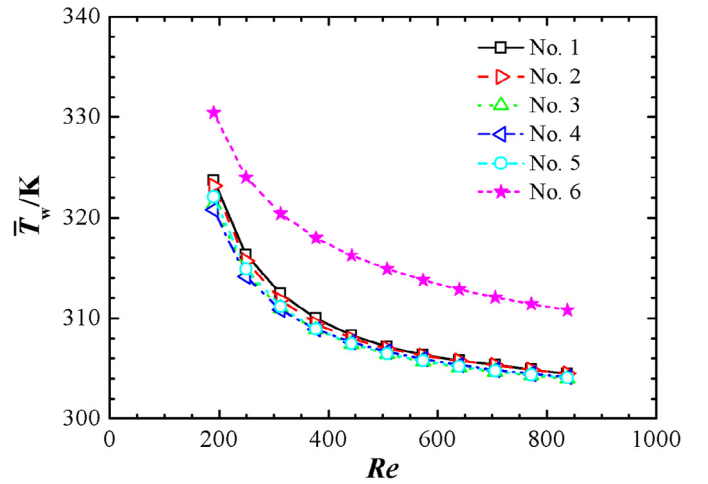


Fig. 8. Variation of $\bar{T}_w$ with Re .

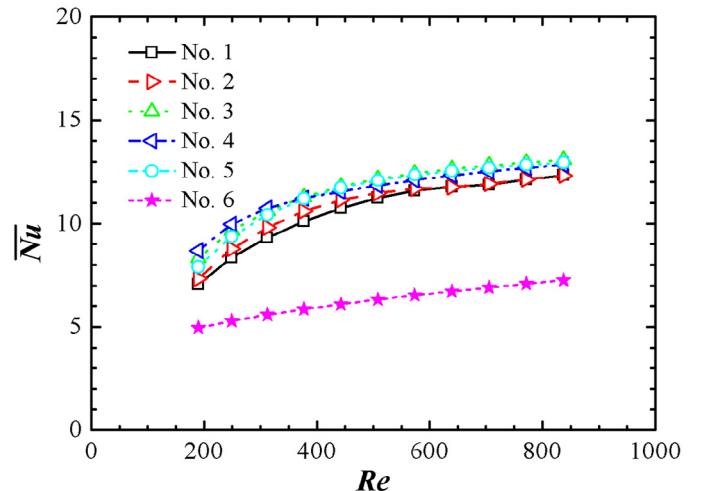


Fig. 9. Variation of $\bar{Nu}$ with Re .

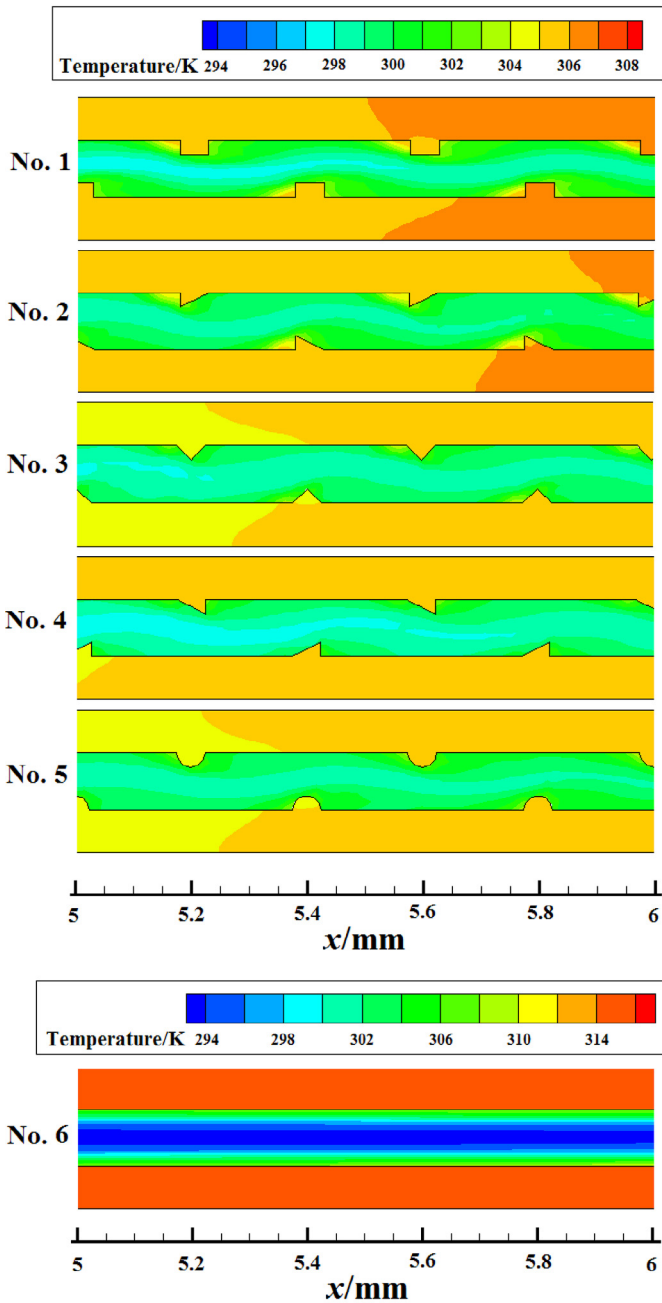


Fig. 10. Temperature distributions in the microchannels ($Re = 508$, $z = 0.25$ mm).

but the difference is not large. As shown in Fig. 9, microchannel heat sinks No. 3, No. 4 and No. 5 have a slightly better heat transfer performance than No.1 and No.2.

In order to further discuss the local heat transfer performance, Fig. 10 shows the temperature distributions in the microchannels with offset ribs at $Re = 508$ in the x - y plane ($z = 0.25$ mm). It can be seen that the temperature of the silicon for the smooth microchannel heat sink is much higher than that in the newly proposed ones, which indicated again the enhancement effect of the offset ribs on heat transfer. It can be observed that the offset ribs lead to a noticeable decrease of temperature difference between the channel center and the sidewalls. This temperature distribution may be a consequence of the effective mixing of the hot water near side-wall and the cold water near the channel center. Besides, it can be also found that corresponding to the recirculation region in the

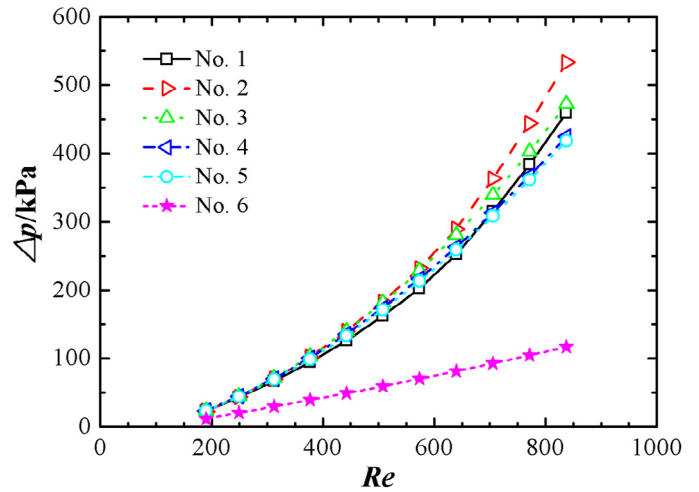


Fig. 11. Variation of Δp with Re .

downstream of the ribs as shown in Fig. 5, the water temperature is extremely similar nearby, and corresponding to the small laminar stagnation zone in the upstream of the ribs for the microchannels No.1 and No.2 as shown in Fig. 6, the temperature of water quite closes to the adjacent sidewall and is obviously higher than the nearby.

3.4. Pressure drop

Figures 11 and 12 respectively show the variation of pressure drop (Δp) and Poiseuille number ($\bar{f}Re$) with Re in the range of 190 to 838. Here, the apparent friction factor is used to calculate the Poiseuille number to account for the pressure drop due to friction and the developing region effects, which represents an average value of the friction factor over the flow length [26]. It is noted that Δp and $\bar{f}Re$ show a sustained and rapid increase with increase of Re for the newly proposed heat sink. Further, different offset ribs result in different increase rates of $\bar{f}Re$. Microchannels No.1 and No.2 show the most rapid increase rate, No.3 comes second, and No.4 and No.5 show the slowest. For microchannels No.3, No.4 and No.5, $\bar{f}Re$ increases approximately linearly with Re and the gradient for microchannel No.3 is larger than the other two.

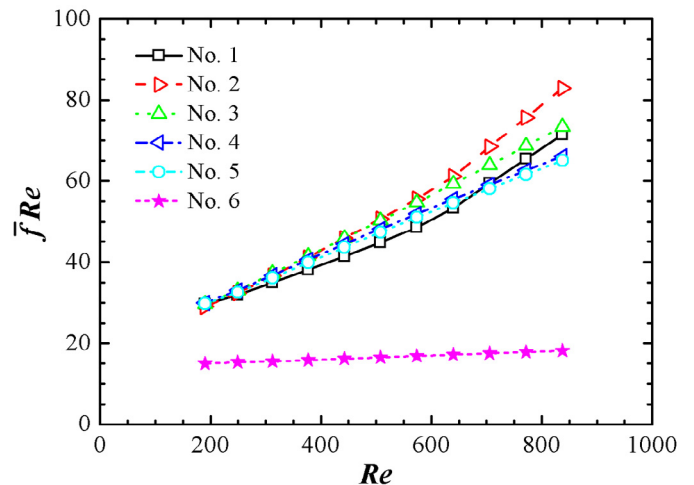


Fig. 12. Variation of $\bar{f}Re$ with Re .

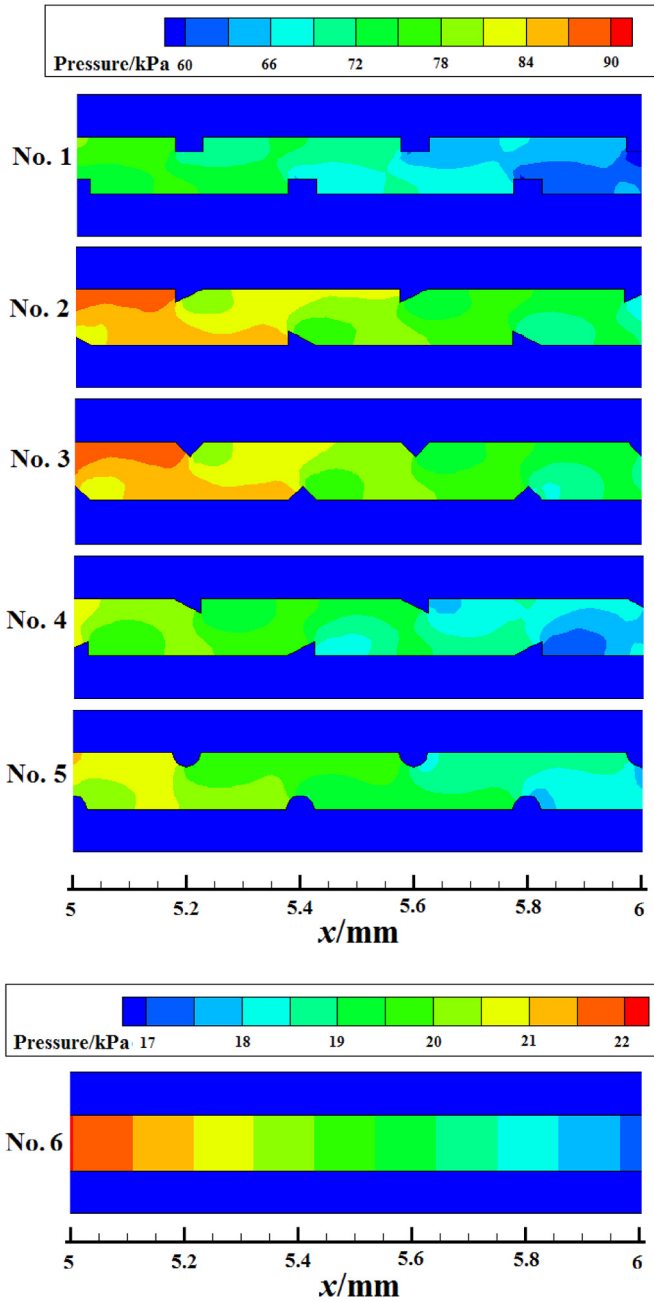


Fig. 13. Pressure distributions in the microchannels ($Re = 508$, $z = 0.25$ mm).

Figure 13 presents the corresponding pressure distribution at $Re = 508$ in the x - y plane ($z = 0.25$ mm). For the smooth microchannel, the pressure continuously decreases due to frictional losses along the microchannel, and for the microchannel with offset ribs, the pressure drop consists of not only the term along the microchannel, but also the term resulting from offset ribs. When water flows across the offset ribs, the velocity increases and the pressure decreases with the constriction of flow passage; while water flows over the offset ribs, the velocity decreases and the pressure increases with the expansion of flow passage. In general, the pressure generally drops tortuously with periodical fluctuations for the microchannels with offset ribs. It is noted that for the microchannels with offset ribs, the pressure at the recirculation is lower than the vicinity and the pressure near the opposite sidewall is obviously higher. The rever-

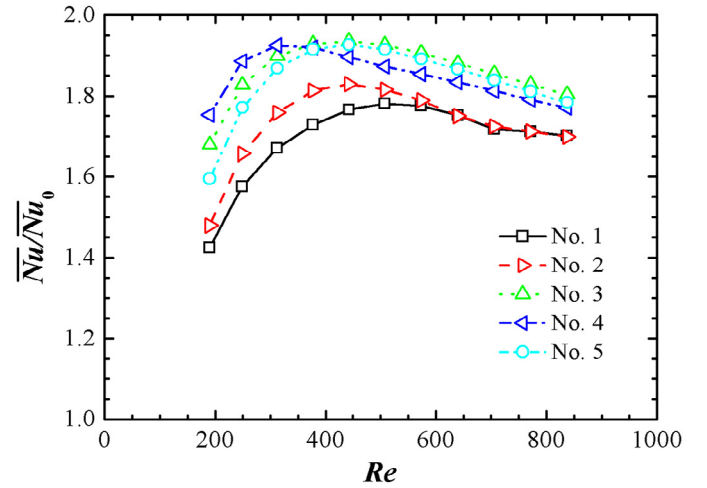


Fig. 14. Variation of $\overline{Nu}/\overline{Nu}_0$ with Re .

sal pressure gradient results in the formation of the recirculation and thus the effective mixing of hot and cold water.

3.5. Performance evaluation

In order to further evaluate the heat transfer performance and compare with the smooth microchannel heat sink, Figs. 14, 15 and 16 show the variations of $\overline{Nu}/\overline{Nu}_0$, $\overline{f}/\overline{f}_0$ and PEC with Re in the range of 190 to 838, respectively. As one can be seen from Figs. 14 and 15, the newly proposed microchannel heat sinks yield Nusselt number of 1.42–1.95 and apparent friction factor of 1.93–4.57 times higher than the smooth one. As shown in Fig. 14, for microchannel heat sinks No.1, No.2, No.3 and No.5, $\overline{Nu}/\overline{Nu}_0$ tends to increase for $Re < 400$, reaches a maximum when $400 < Re < 500$ and then drops with further increase of Re . For microchannel heat sink No.4, $\overline{Nu}/\overline{Nu}_0$ reaches the maximum more early than the others at about $Re = 300$. As shown in Fig. 15, for the newly proposed microchannels, $\overline{f}/\overline{f}_0$ rises rapidly with the increase of Re . This different tendency of $\overline{Nu}/\overline{Nu}_0$ and $\overline{f}/\overline{f}_0$ with Re leads to deterioration of PEC at higher Re as shown in Fig. 16. It is noticeable that as Re increases, PEC increases greatly at first, reaches a maximum and then decreases rapidly. As a consequence of serious pressure drop, the newly proposed microchannel heat sinks gradually lose their advantages as an effective heat transfer enhancement method at higher Re . Additionally, different offset ribs result in different Re corresponding to the largest PEC . Microchannel heat sinks No.4 and No.3 are the first two to reach a maximum at about $Re = 240$, No.2 and No.5 come second at about $Re = 350$, and No.1 is the last at about $Re = 450$. However, the newly proposed microchannel heat sinks still yield PEC of 1.02–1.48, despite lower than $\overline{Nu}/\overline{Nu}_0$. As $Re < 350$, No.4 yields the highest PEC while No.1 shows the lowest, while as $Re > 400$, No.5 yields the highest PEC while No.2 shows the lowest.

4. Conclusions

Laminar flow and heat transfer in microchannel heat sinks with offset ribs on sidewalls have been investigated numerically. The three-dimensional equations considering entrance effect, conjugate heat transfer, viscous heating and temperature-dependent thermo-properties are solved the fluid flow and heat transfer. From the present work, the following conclusions can be drawn:

- (1) The average Nusselt number for the newly proposed microchannel heat sinks are higher than the smooth one, but

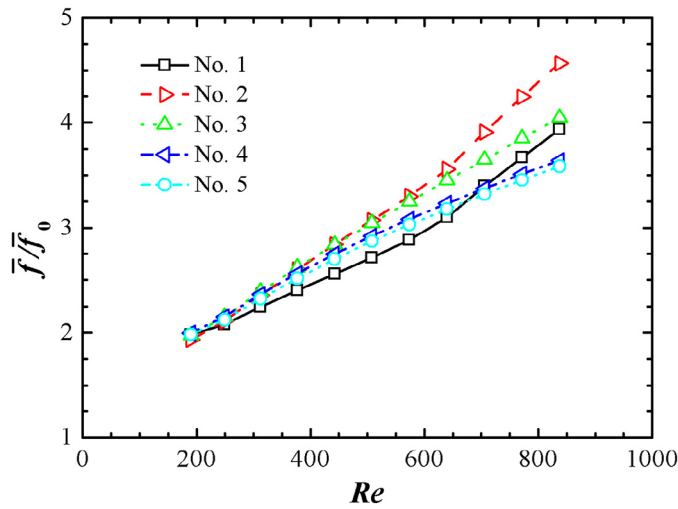


Fig. 15. Variation of $\bar{f}/\bar{f}_0$ with Re .

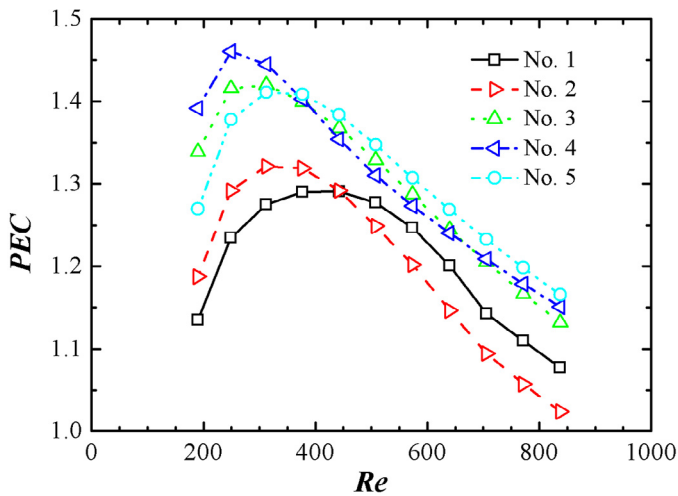


Fig. 16. Variation of PEC with Re .

their increase rates slow down as Reynolds number increased, while the Poiseuille number tend to increase continually.

- (2) Depending on the different offset ribs and Reynolds number ($190 \leq Re \leq 838$), the newly proposed microchannel heat sinks yield Nusselt number of 1.42–1.95 and apparent friction factor of 1.93–4.57 times higher than the smooth one, leading to the performance evaluation criteria of 1.02–1.48.
- (3) As $Re < 350$, the microchannel heat sink with forward triangular offset ribs yields the highest performance evaluation criteria while the one with rectangular offset ribs shows the lowest. As $Re > 400$, the one with semicircular offset ribs brings about the best performance evaluation criteria while the one with backward triangular offset ribs gives the worst.

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Nomenclature

A	Area
c_p	Specific heat capacity
D_h	Hydraulic diameter
f	Fanning friction factor
$\bar{f}$	Average Fanning friction factor
$\bar{f}_0$	Average Fanning friction factor for the smooth microchannel
h	Heat transfer coefficient
$\bar{h}$	Average heat transfer coefficient
$\bar{h}_0$	Average heat transfer coefficient for the smooth microchannel heat sink
H	Height of computational domain
H_{mic}	Height of the microchannel
k	Thermal conductivity
L	Length of computational domain
n	Local coordinate normal to the wall
Nu	Nusselt number, hD_h/k_f
$\bar{Nu}$	Average Nusselt number
$\bar{Nu}_0$	Average Nusselt number for the smooth microchannel heat sink
p	Pressure
PEC	Performance evaluation criteria
q	Heat flux
Re	Reynolds number, $\rho_f u_m D_h / \mu_f$
T	Temperature
u, v, w	Velocity components in x, y and z directions
$\dot{V}$	Volumetric flow rate
W	Width of computational domain
W_{mic}	Width of microchannel
Δp	Pressure drop
x, y, z	Cartesian coordinates

Greek letters

ρ	Density
μ	Dynamic viscosity

Subscripts

con	Contact
f	Fluid
in	Inlet
m	Mean
mic	Microchannel
out	Outlet
pp	Pumping power
s	Solid
w	Silicon base

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